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UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

UIUC STATISTICS DATATHON

MARCH 27 – APRIL 1, 2026

Your Datathon Hosts

Sid Chakravarty

VP, Business Intelligence, Performance Insights

Connect with Me!

Ryan McMahon

AVP, Operations Planning and Analysis

Connect with Me!

Sal Pepe

AVP, Advanced Insights & Analytics

Connect with Me!

SYNCHRONY OVERVIEW

FORTUNE 500

Synchrony is the **largest private label credit card company in the U.S.** based on purchase volume and receivables.

WHAT WE DO

We're a **premier consumer financial services company** delivering **customized financing programs** across key industries including retail, health, auto, travel and home. Along with **award-winning consumer banking products**, we bring deep industry expertise, actionable data insights, innovative solutions and differentiated digital experiences to improve the success of every business we serve and the quality of each life we touch.

90+

YEARS OF HISTORY

Built One Customer At A Time

70M+

ACTIVE CUSTOMERS

Accounts Across Multiple Industries

20K+

EMPLOYEES

Each One Integral To Our Business

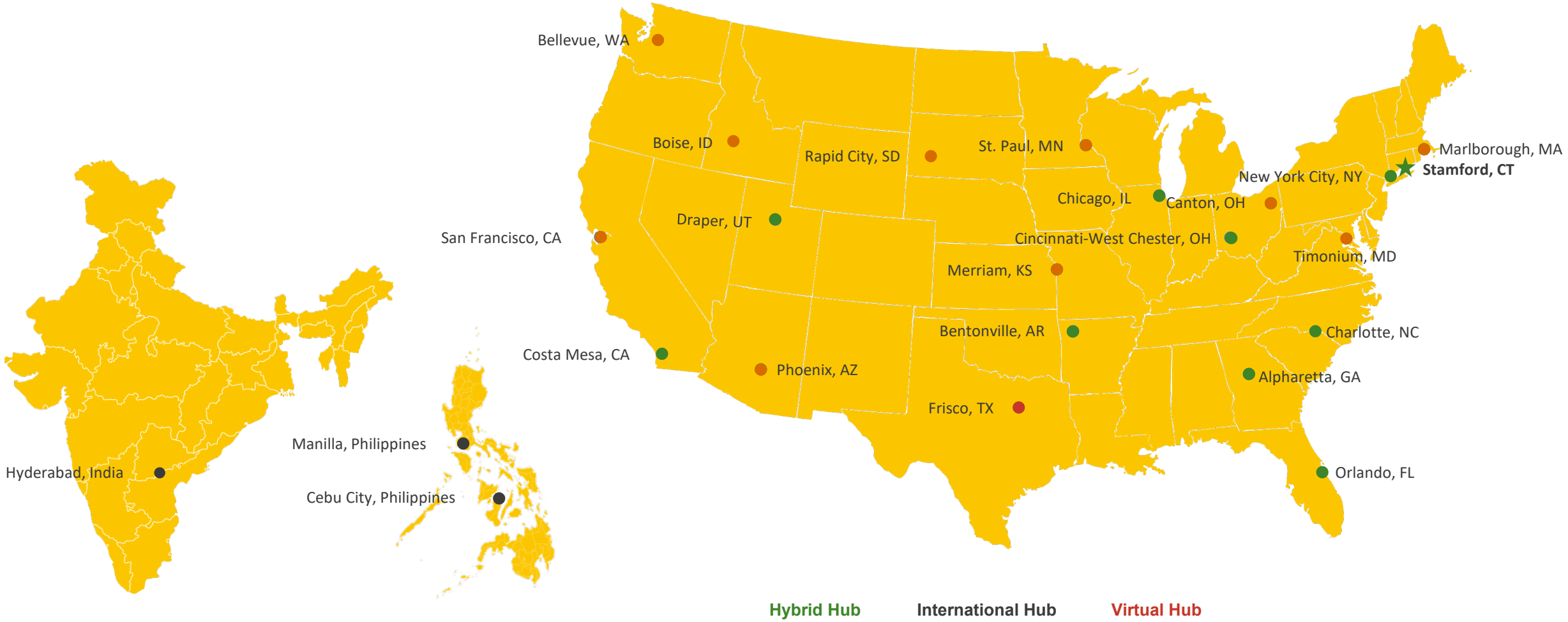
PLATFORMS & PRODUCTS

Synchrony has and is continuing to build a robust portfolio of consumer financial products that positively impact consumers' lives in multiple, meaningful ways.

DIGITAL	venmo, verizon
HEALTH & WELLNESS	Pets Best PET HEALTH INSURANCE, Walgreens, CareCredit℠
HOME & AUTO	LOWE'S, Ashley HOMESTORE
DIVERSIFIED & VALUE	TJX, sam's club
LIFESTYLE	Kawasaki, jtv jewelry love

SYNCHRONY HUBS

We have over **20,000 employees worldwide** including the **United States**, **India**, and the **Philippines**. Our global presence allows us to engage with our partners and customers when they want, building strong relationships and enabling growth.



2026 DATATHON OVERVIEW

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ILLINOIS

powered by aws

databricks

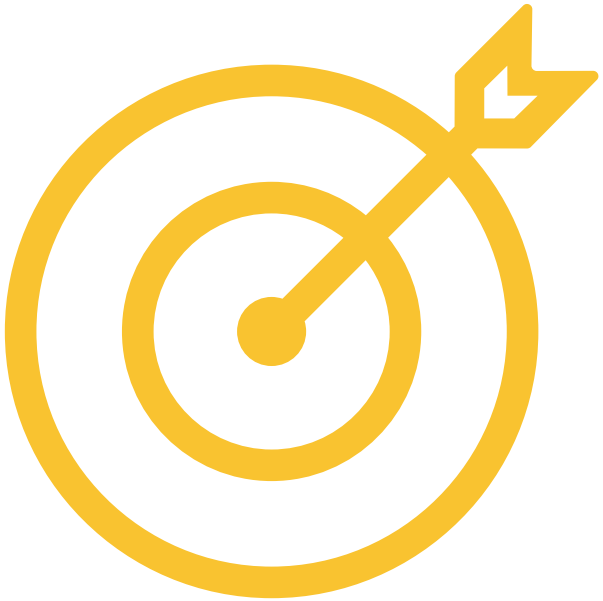
What Workforce Planning Does

Workforce Planning is responsible for ensuring our business is properly staffed to service our customers and clients across multiple channels. Our organization handles many tasks including forecasting, capacity planning, scheduling, real-time management, reporting, and more.

Why Forecast Accuracy is Critical

Our agents handle over 4 million calls each month. We have dozens of contractual commitments with our partners that are tied to call center servicing. Forecast accuracy is extremely important to our business because it helps us ensure we:

- Are being proactive and responsible with hiring.
- Schedule agents to align to call volume.
- Hit our budget targets.
- Meet all our contractual commitments.
- Have a strong tactical plan.



Objective

Develop a model to predict intraday contact center inbound call volume at the portfolio level. Forecast data should account for 24/7 servicing in 30-minute intervals for each day in a month. Historical data at daily & interval levels has been provided for 4 portfolios of varying sizes within the Diversified & Value platform.

Model should be designed to predict the following metrics in 30-minute intervals:

- Call Volume (CV)
- Customer Care Time (CCT)
- Abandon Rate (ABD)

*All interval data should be in EST

Operational Requirements

- Model should predict all metrics at every 30-minute interval for each day in a month.
- Model needs to scale to portfolios of different sizes, longer/shorter AHT, different SL targets.
- Interval data needs to account for intraday, intraweek, intramonth patterns.
- Model should be able to account for seasonality changes:
  - Daylight Savings time, holiday sales events, etc.
- Approach should be explainable and repeatable.
- Forecasted values **cannot** be negative (applies to volume, CCT, Abandoned Rate)
- **Model Output Template** is provided through the Box
  - Forecasted values must align with rows and columns provided

Deliverables & Rules

There are 3 deliverables for This Datathon:

- Developed Code & Presentation with 7-Minute-Video uploaded to the Box

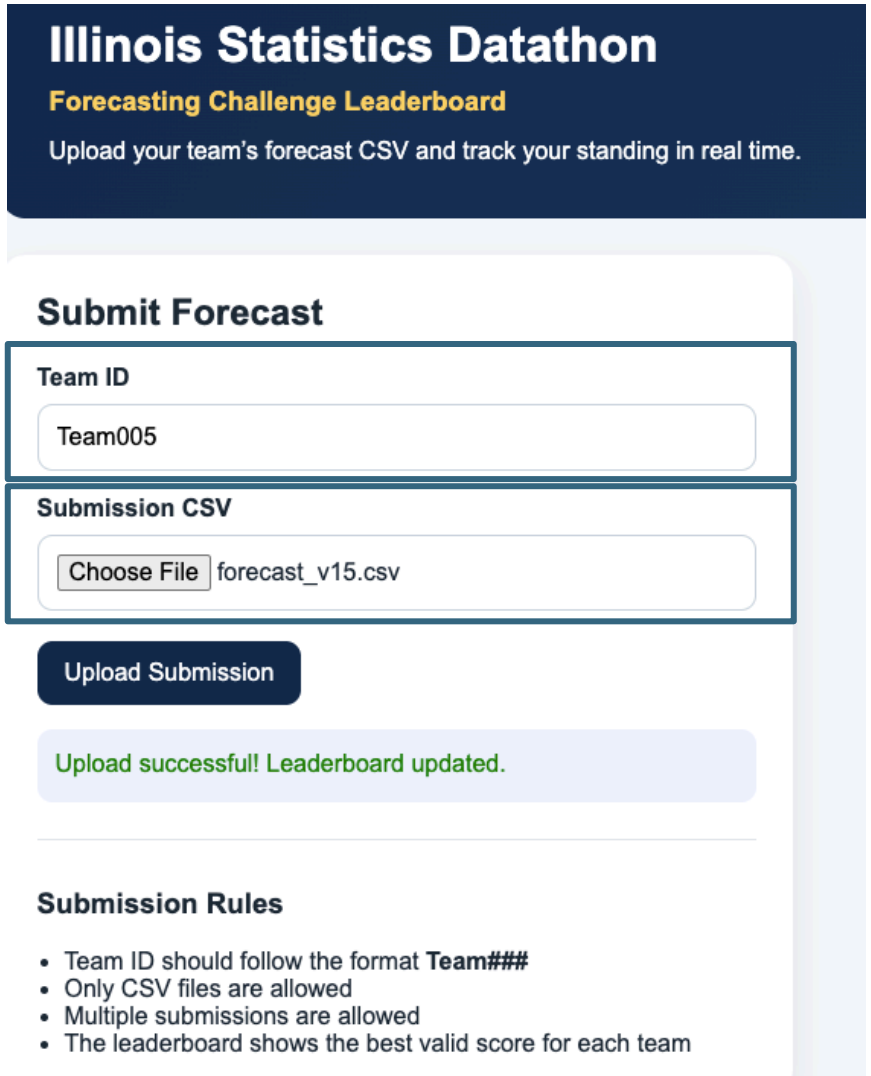
- Model Output uploaded to AWS

Rules for This Datathon:

- All submissions must be in by **12:00 PM on Sunday (3/29)**, there will be no exceptions.
- All submissions must have a **7-minute video, presentation and zipped code** as deliverables, or they won't be counted.
- No use of Generative AI.
- Additional guidelines for each deliverable in following slides.

Model Output Submission Guidelines

- Check your email for your assigned *Team ID*
- Open this link to AWS: here
- Enter your *Team ID*
  - Formatted as Team###: ID 1 = Team001, ID 10 = Team010, ID 100 = Team100
- Upload your output file as a .csv
  - Format filename as forecast_v##.csv: version 1 = forecast_v01.csv
  - Teams can upload multiple forecast versions, but all versions must be aligned to the same Team ID
  - The same file cannot be uploaded twice under the same Team ID
    - Ex: Team001 cannot upload forecast_v01.csv twice, instead must upload forecast_v01 & forecast_v02
- Click "Upload Submission"



Code Submission Guidelines

Do's

- Build a baseline model first. Add complexity and evaluate performance iteratively.
- All design, development, and testing must be done inside approved Docker environments. Three Docker images will be made available to students to choose.
  - PyTorch / Tensorflow / Classifical ML + Boosting (e.g., Scikit Learn, Statsmodels, pmdarima, XGBoost, Prophet)
- Feel free to perform feature engineering to detect features, such as day-of-week, holidays, etc.
- Include a README file with clear instructions on setup of the code, data used and commands to run code. Include the link to the video in the README file.
- Code should be zipped and uploaded to the project folder allocated for your team in the box.

Don't's

- Students are not allowed to use any Generative AI tool, either directly or indirectly to design, develop, and present. All code and artifacts produced must be human-written. Any use of Generative AI will lead to disqualification.
- Do not install external packages.
- Do not overfit the data, especially the 3-month intra-day data.
- Don't ignore negative bias (i.e., understaffing).

Understanding Cost-Sensitive and Asymmetric Leaderboard Scores

Overall Composite Score: $(\omega_1 \times E_V) + (\omega_2 \times E_C) + (\omega_3 \times E_B) + (\omega_4 \times P_t)$

Error$_{Volume}$(E_V): $\frac{\sum_{t=1}^{T} | A_t^{(V)} - F_t^{(V)} |}{\sum_{t=1}^{T} A_t^{(V)}}$; *where superscript (V) denotes volume*

Error$_{CCT}$(E_C): $\frac{\frac{1}{T'} \sum_{t \in T^+} | A_t^{(C)} - F_t^{(C)} |}{\frac{1}{T'} \sum_{t \in T^+} A_t^{(C)}}$; *where* $T^+ = \left\{ t : A_t^{(C)} > 0 \right\}$ & $T' = | T^+ |$; *superscript (C) denotes CCT/AHT*

Error$_{Abandon_Rate}$(E_B)[1,2]: $\frac{\sum_{t=1}^{T} | A_t^{(B)} - F_t^{(B)} |}{\sum_{t=1}^{T} A_t^{(B)}}$; *where superscript (B) denotes abandoned*

Penalty$_{Workload}$(P_t): $Penalty_t = \alpha \times \max\left(A_t^{(w)} - F_t^{(w)}, 0\right) + \beta \times \max\left(F_t^{(w)} - A_t^{(w)}, 0\right)$; *where* $W_t = V_t \times C_t$

$$Penalty_{Workload} = \frac{\sum_{t \in T^+} Penalty_t}{\sum_{t \in T^+} A_t^{(w)}}$$

Additional Notes:

1. Abandoned Calls = Abandoned Rate x Calls Offered
2. When submitting, include both Abandoned Calls and Abandoned Rate

Datathon Timeline

Friday

Friday 3/27:

Opening Ceremony – 5PM CST

Datathon Begins – 6PM CST

Saturday / Sunday

Saturday 3/28:

AWS Office Hours – 9AM CST (1 hr)

Synchrony Office Hours – 11AM CST (1 hr)

Synchrony Office Hours – 3PM CST (1 hr)

AWS Office Hours – 4PM CST (1 hr)

Sunday 3/29:

Datathon Ends – 12PM CST

Monday / Tuesday / Wednesday

Monday & Tuesday:

Judging & Finalists are notified via Email

Wednesday 4/1:

Finalists Present & Winner is Announced – 7PM CST

Model Output Template

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Month	Day	Interval	Calls_Offered_A	Abandoned_Calls_A	Abandoned_Rate_A	CCT_A	Calls_Offered_B	Abandoned_Calls_B	Abandoned_Rate_B	CCT_B	Calls_Offered_C	Abandoned_Calls_C	Abandoned_Rate_C	CCT_C	Calls_Offered_D	Abandoned_Calls_D	Abandoned_Rate_D	CCT_D
2	August	1	0:00																
3	August	1	0:30																
4	August	1	1:00																
5	August	1	1:30																
6	August	1	2:00																
7	August	1	2:30																
8	August	1	3:00																
9	August	1	3:30																
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33	August	1	15:30																
34	August	1	16:00																
35	August	1	16:30																
36	August	1	17:00																

- Output template is loaded into the Box, you will need to have your model output populated into the empty cells

Helpful Resources

- Make sure to upload all your files to your team folder in the box: **LINK**

- Link to Data File Folder: **LINK**
  - Training dataset, model output template, PPT, Team ID assignments

- Use the discord channel to post any questions and watch for announcements: **LINK**

- Link to AWS submission page: **LINK**

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PRIZES

2nd Place

Apple Airpods

1st Place

Nintendo Switch

3rd Place

Google Home

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CHANGING WHAT'S POSSIBLE

Data Definitions

- **Interval** - 30-minute period of the day when calls were offered. In 24-hour, Eastern Time. (EX: 13:00 is 1PM-1:30PM).
- **Call Volume (CV)**- Total number of incoming calls (answered + abandoned).
- **Abandoned Calls** - Total number of callers who hung up before call was answered.
- **Abandoned Rate (ABD)** - Percentage of total callers who hung up before call was answered.
- **Customer Care Time (CCT)** - Average time it takes an agent to handle each call. Includes Talk Time, Hold Time, and time notating & actioning account after the call.
- **Daily Staffing** - Total number of Full Time Agents scheduled daily for each servicing group.